Trigonometric Identity Proofs

Precalculus Practice — 45 Minutes

Name: _____ Date: _____

Instructions

Prove each identity by transforming the left-hand side into the right-hand side. Show a chain of equal expressions and name the identity or algebraic rule used at each step. Do not manipulate both sides as an equation or verify with numerical examples. Use exact expressions; no calculator. All angles are in radians. Assume variables take values for which both sides are defined; observe any additional interval restriction.

This set covers cofunction, sum/difference, double-angle, half-angle, and even/odd identities. Basic reciprocal, quotient, and Pythagorean identities may also be used.

Prove each identity

1.

$$\frac{\cos\left(\frac{\pi}{2} - x\right)}{\cos(-x)} = \tan x$$

2.

$$\tan(-x)\csc(-x) = \sec x$$

3.

$$\sin(x + y) + \sin(x - y) = 2\sin x \cos y$$

4.

$$\frac{1 - \cos(2x)}{\sin(2x)} = \tan x$$

Trigonometric Identity Proofs — Continued

5.
$$\frac{\sec\left(\frac{\pi}{2} - x\right)}{\cot x} = \sec x$$

6.
$$\frac{\tan(x+y) - \tan x}{\tan y} = \frac{1 + \tan^2 x}{1 - \tan x \tan y}$$

7.
$$\frac{\sin(-x) + \cos\left(\frac{\pi}{2} + x\right)}{\cos(-x)} = -2\tan x$$

8.
$$\frac{\cos(2x)}{\cos x - \sin x} = \cos x + \sin x$$

Trigonometric Identity Proofs — Continued

9. $$\frac{\sin^2\left(\frac{x}{2}\right)}{1+\cos x} = \frac{1}{2}\tan^2\left(\frac{x}{2}\right)$$

10. For $0 < x < \pi$,
$$\sqrt{\frac{1-\cos x}{1+\cos x}} = \tan\left(\frac{x}{2}\right).$$

11. $$\frac{\sin(3x)-\sin x}{\cos(3x)+\cos x} = \tan x$$

12. $$\frac{\sin(x+y)}{\sin(x-y)} = \frac{\tan x+\tan y}{\tan x-\tan y}$$